Luke Zambella

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Employment

Systems Developer III *Verizon, 1 Verizon Way, Basking Ridge NJ, 07920* June 2021 – Present

- Head developer for a Spring Boot backend that interprets requests from a React frontend and integrates with numerous backend systems to provide customer service agents generative AI functionality such as real-time Q&A, voice call summarization, and call analytics for business analysis.
- Supported a second Spring Boot application responsible for processing millions of packets of live call data per day to big data and pushing real time transcriptions to customer service agents over numerous support channels.
- Provided APIs for real-time IVR agent supervisor monitoring that alerts supervisors based on IVR call customer sentiment, summarizes the conversation, provides live transcript and sentiment monitoring, and gives the ability to escalate to a live customer support agent.
- Created a set of APIs that provides an agent with an AI generated post-call summarization along with agent-to-agent transfer reasons from real time call transcription that can be manually edited. Expanded upon the implementation to different support channels based upon their own requirements like additional points of summarization.
- Implemented a system for generative AI answers from real time call transcription in order to decrease average call time through providing in-context answers utilizing customer data using Google Cloud.
- Leveraged Gemini using prompt engineering to extract useful data (contact information, dollar amount, sim/IMEI number) from real-time transcripts then pushed the extracted data into a convenient format for agent's dashboard to reduce average call time by minimizing the amount of times an agent may ask a customer for data.
- Implemented Gemini once more with prompt engineering to analyze live transcriptions to determine customer intent to cancel service and automatically push retention offers to an agent who can provide to the customer.
- Designed a method for tracking an agent's chatbot usage to determine whether to show a request for feedback. Stored agent usage details in a Postgres table using Redis cache to minimize database calls and stored feedback in another table. Created RESTful APIs that can be used to retrieve and analyze feedback.

Projects

- **PID Controller for Espresso Machine (C):** Designed the schematic for a custom PID controller for precisely adjusting water temperature for a home espresso machine. Prototyped design with an AVR development board and implemented firmware with C. Ensured that the design was safe by adding multiple killswitches. Worked with a fabricator to manufacture prototype PCBs. Iterated on design by addressing original shortcomings and new features to the next one.
- **Custom Keyboard (C):** Used KiCad to design the schematic and PCB of a USB HID keyboard. Implemented firmware using an AVR MCU with open source QMK as a base. Worked with a PCB fabricator to build prototype PCBS. Designed keycaps and the case using Autodesk Fusion and rapidly tested design with 3d manufacturing. Iterated upon design by analyzing shortcomings of the previous prototypes and making improvements.
- **Checkers Playing Robot (Python):** Developed a checkers game utilizing computer vision to showcase robot interaction with physical objects. The robot was able to discern the game board from a camera feed and determine the game's state. Integrated a checkers engine to provide interactivity between the robot and the player.
- **Home server:** Deployed a Ubuntu Linux image to a server to host applications using Docker Compose. Set up a personal VPN using WireGuard for secure remote access. Setup custom domain with DNS A name routing through cloudflare for subdomains to expose specific applications to others using nginx reverse proxy.

Education

The College of New Jersey (TCNJ), *Ewing NJ:* Bachelor of Science (B.S.) in Computer Engineering *3.3/4.0 GPA* May *2021*

Coursework: *Computer Architecture and Organization, Electronics, Control Systems, Software Engineering, Artificial Intelligence (AI), Natural Language Processing (NLP), Digital Signal Processing (DSP)*

Skills

Programming Languages: Java, Python, Javascript, C#, C, C++

Frameworks: Spring, .NET Core, React.js, Node.js, Bootstrap, Qt

Software: Jenkins, Docker, Docker Compose, Postman,, Linux, Microsoft Office, Google Suite, Cursor AI